





DATA 300

Predicting Flight Arrival Delays for Flights Departing from Harrisburg International Airport (MDT)

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PROJECT & DATA OVERVIEW

Project overview

- Predict whether flights from MDT arrive ≥15 minutes delayed
- Motivation: improve scheduling, resource allocation, and passenger experience
- Approach: use historical flight data + machine learning classification

Data overview

- BTS On-Time Performance dataset, filtered to MDT departures (~29,000 flights)
- Each row = one flight
- Target: ARR_DEL15 (1 = delayed, 0 = on-time)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	YEAR	QUARTER	MONTH	DAY_OF_MO	DAY_OF_WE	FL_DATE	OP_UNIQUE_	OP_CARRIER	OP_CARRIER	TAIL_NUM	OP_CARRIER	ORIGIN_AIRP	ORIGIN_AIRP	ORIGIN_CITY	ORIGIN	ORIGIN_CITY	ORIGIN_STAT	ORIGIN_STAT	ORIGIN_STAT	ORIGIN_WAC	DEST_AIRPOI	DEST_AIRPOI	DEST_CITY_M	DEST	DEST_CITY_N	DEST_STATE_
2	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N301PQ	5184	14100	1410005	34100	PHL	Philadelphia	PA	42	Pennsylvania	23	13487	1348702	31650	MSP	Minneapolis	MN
3	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N301PQ	5222	11433	1143302	31295	DTW	Detroit, MI	MI	26	Michigan	43	14100	1410005	34100	PHL	Philadelphia	PA
4	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N305PQ	4786	10135	1013506	30135	ABE	Allentown/Be	PA	42	Pennsylvania	23	10397	1039707	30397	ATL	Atlanta, GA	GA
5	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N307PQ	4823	11433	1143302	31295	DTW	Detroit, MI	MI	26	Michigan	43	14122	1412202	30198	PIT	Pittsburgh, P	PA
6	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N307PQ	4855	14122	1412202	30198	PIT	Pittsburgh, P	PA	42	Pennsylvania	23	10721	1072102	30721	BOS	Boston, MA	MA
7	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N325PQ	5253	10397	1039707	30397	ATL	Atlanta, GA	GA	13	Georgia	34	10135	1013506	30135	ABE	Allentown/Be	PA
8	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N367CA	5059	10397	1039707	30397	ATL	Atlanta, GA	GA	13	Georgia	34	13230	1323002	32070	MDT	Harrisburg, P	PA
9	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N367CA	5059	13230	1323002	32070	MDT	Harrisburg, P	PA	42	Pennsylvania	23	10397	1039707	30397	ATL	Atlanta, GA	GA
10	2020	1	1	1	3	1/1/20 0:00	9E	20363	9E	N478PX	4778	13487	1348702	31650	MSP	Minneapolis	MN	27	Minnesota	63	14122	1412202	30198	PIT	Pittsburgh, P	PA

	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ
1	DEST_STATE_	DEST_STATE_	DEST_WAC	CRS_DEP_TI	DEP_TIME	DEP_DELAY	DEP_DELAY_	DEP_DEL15	DEP_DELAY_	DEP_TIME_BL	TAXI_OUT	WHEELS_OF	WHEELS_ON	TAXI_IN	CRS_ARR_TI	ARR_TIME	ARR_DELAY	ARR_DELAY_	ARR_DEL15	ARR_DELAY_	ARR_TIME_BL	CANCELLED	CANCELLATI	DIVERTED	CRS_ELAPSE	ACTUAL_ELA
2	27	Minnesota	63	1640	1638	-2	0	0	-1	1600-1659	11	1649	1821	7	1850	1828	-22	0	0	-2	1800-1859	0		0	190	170
3	42	Pennsylvania	23	1230	1230	0	0	0	0	1200-1259	18	1248	1354	4	1413	1358	-15	0	0	-1	1400-1459	0		0	103	88
4	13	Georgia	34	0600	0555	-5	0	0	-1	0600-0659	9	0604	0759	4	[illegible]	0803	-30	0	0	-2	0800-0859	0		0	153	128
5	42	Pennsylvania	23	1235	1225	-10	0	0	-1	1200-1259	24	1249	1325	5	1348	1330	-18	0	0	-2	1300-1359	0		0	73	65
6	25	Massachuse	13	1510	1505	-5	0	0	-1	1500-1559	10	1515	1630	10	1652	1640	-12	0	0	-1	1600-1659	0		0	102	95
7	42	Pennsylvania	23	1745	1743	-2	0	0	-1	1700-1759	14	1757	1929	4	1950	1933	-17	0	0	-2	1900-1959	0		0	125	110
8	42	Pennsylvania	23	0940	0935	-5	0	0	-1	0900-0959	35	1010	1133	5	1133	1138	5	5	0	0	1100-1159	0		0	113	123
9	13	Georgia	34	1208	1205	-3	0	0	-1	1200-1259	12	1217	1357	7	1419	1404	-15	0	0	-1	1400-1459	0		0	131	119
10	42	Pennsylvania	23	0915	0920	5	5	0	0	0900-0959	30	0950	1220	5	1230	1225	-5	0	0	-1	1200-1259	0		0	135	125

	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM
1	AIR_TIME	FLIGHTS	DISTANCE	DISTANCE_G	CARRIER_DE	WEATHER_D	NAS_DELAY	SECURITY_D	LATE_AIRCRA	FIRST_DEP_T	TOTAL_ADD_	LONGEST_ADD_GTIME	
2	152	1	980	4									
3	66	1	453	2									
4	115	1	692	3									
5	36	1	201	1									
6	75	1	496	2									
7	92	1	692	3									
8	83	1	620	3									
9	100	1	620	3									
10	90	1	726	3									

Janurary 2020,
all features,
first 10 rows

KEY VARIABLES & FEATURE ENGINEERING





RAW VARIABLES

Total of features: 12
Time:

- day-of-week, month, day-of-month, scheduled departure time

Flight info:

- carrier (OP_UNIQUE_CARRIER)
- destination (DEST)
- distance (DISTANCE)
- elapsed time (CRS_ELAPSED_TIME)

FEATURE ENGINEERING

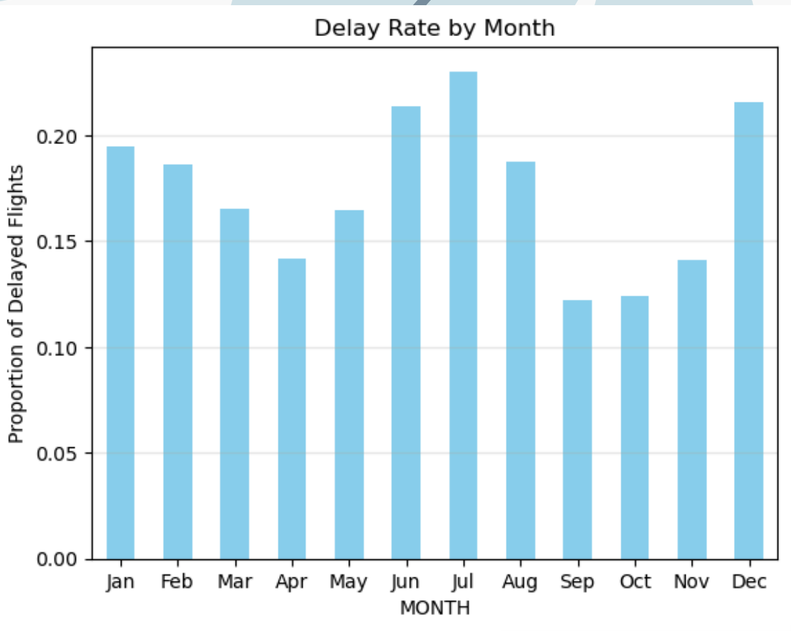
- Converted CRS_DEP_TIME → DEP_HOUR
  → captures time-of-day patterns
- Created PREV_FLIGHT_DELAYED
  → indicates if previous flight was delayed

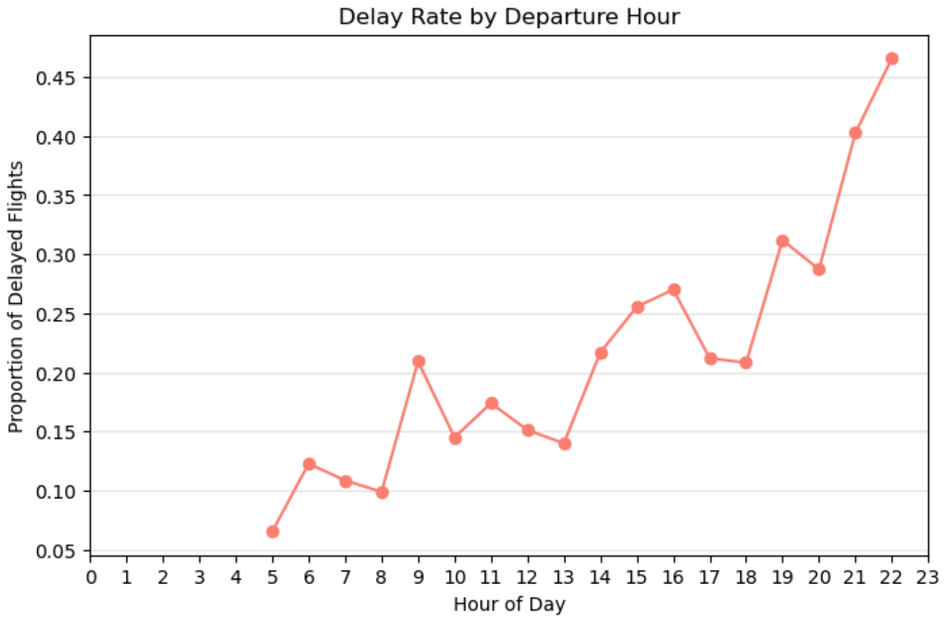
WHY THESE FEATURES

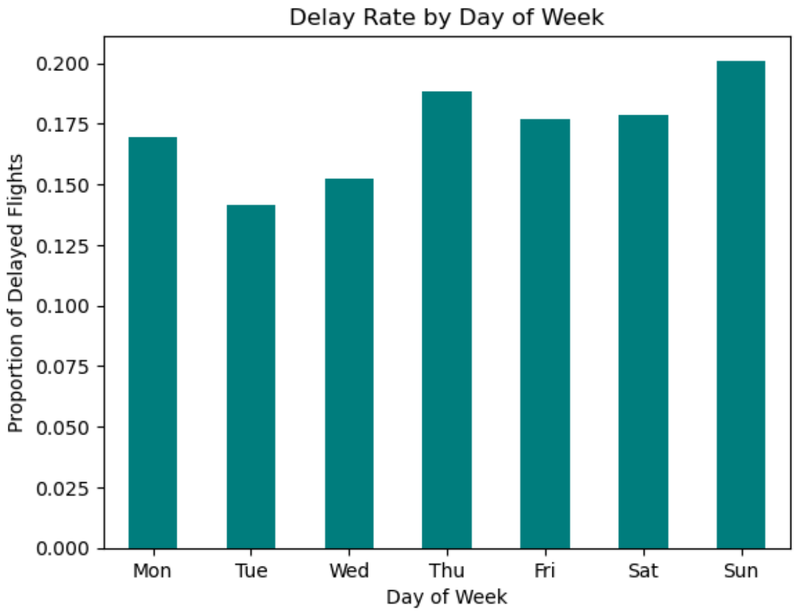
- Time variables capture daily and seasonal delay patterns
- Carrier and destination reflect operational differences across routes
- PREV_FLIGHT_DELAYED captures delay propagation, a key driver of delays

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TIME-BASED PATTERNS







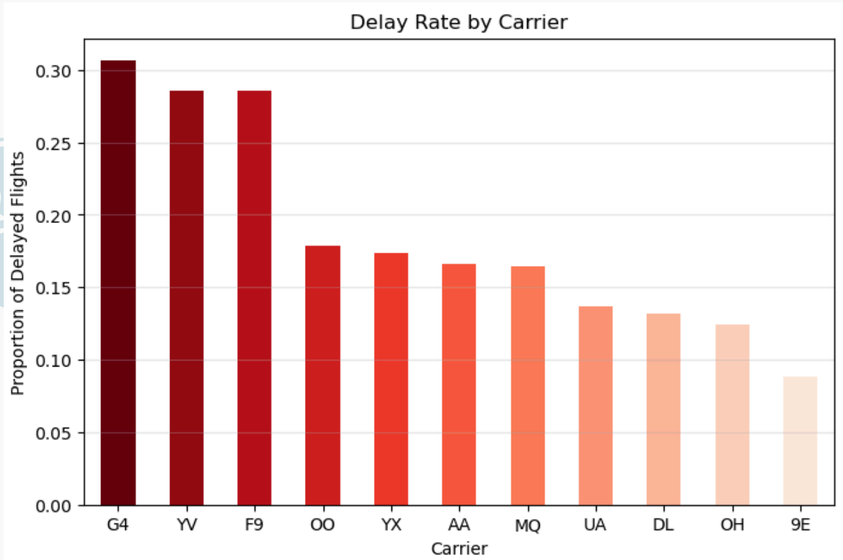
Delays peak in summer (June -July) and December, and are lowest in early fall
→ Suggests seasonal effects (travel demand, weather)

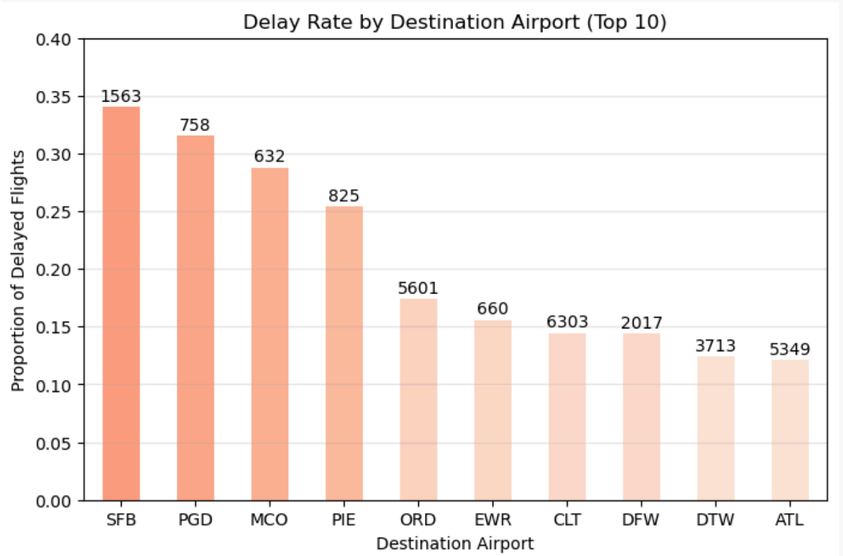
Delays increase steadily throughout the day, with the highest rates at night
→ Suggests delays accumulate across flights during the day

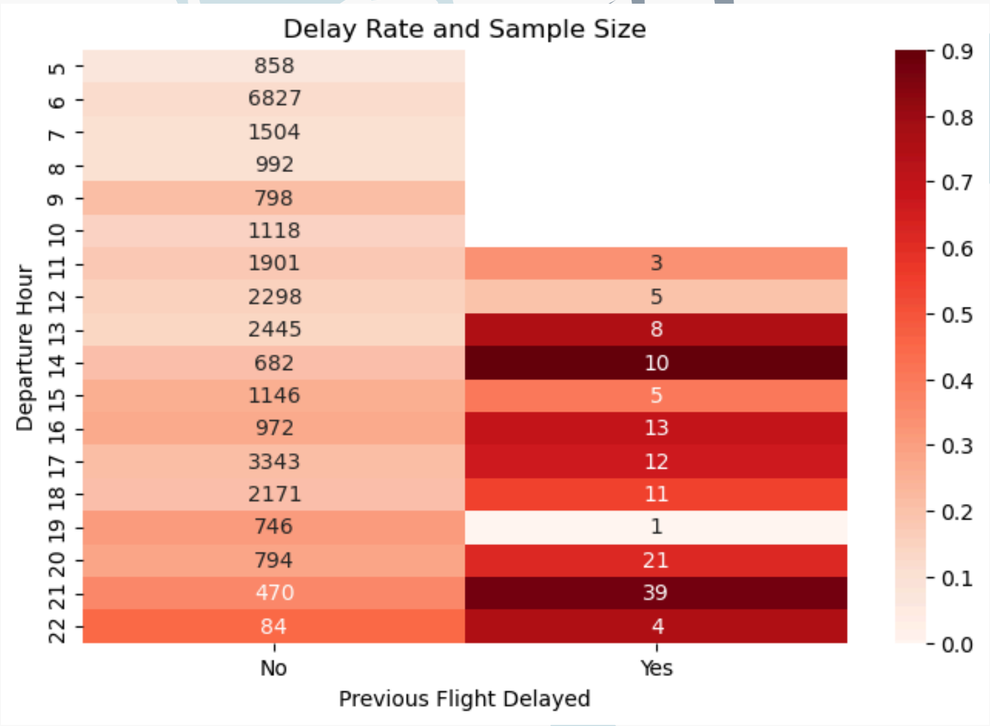
Delays are slightly higher toward the end of the week and weekends (Thu - Sun)
→ Suggests higher traffic and congestion later in the week

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OPERATIONAL FACTORS







ARR_DEL15	0	1
PREV_FLIGHT_DELAYED		
0	0.828433	0.171567
1	0.303030	0.696970

Previous Flight Delay (Cross-tab)

- If previous flight is not delayed → only ~17% chance of delay
- If previous flight is delayed → ~70% chance of delay
- Shows strong delay propagation effect

Delay Rate by Hour & Previous Delay (Heatmap)

- Delays increase later in the day, especially after afternoon/evening
- When previous flight is delayed, delay rates become much higher across all hours
- Combines time effect + propagation → delays compound over the day

MODELING APPROACH



- Binary classification: predict delay (≥15 min)
- Models used: Logistic Regression and Random Forest
- Train/test split for evaluation
- Metrics: Precision, Recall, F1 score
- Focus on recall to minimize missed delays

Why these models

- Logistic Regression: simple, interpretable baseline
- Random Forest: captures nonlinear patterns and interactions

```
# Class balance
df["ARR_DEL15"].value_counts()
df["ARR_DEL15"].value_counts(normalize=True)
```

✓ 0.0s

```
ARR_DEL15
0    0.826065
1    0.173935
Name: proportion, dtype: float64
```

On-Time flights: 83%

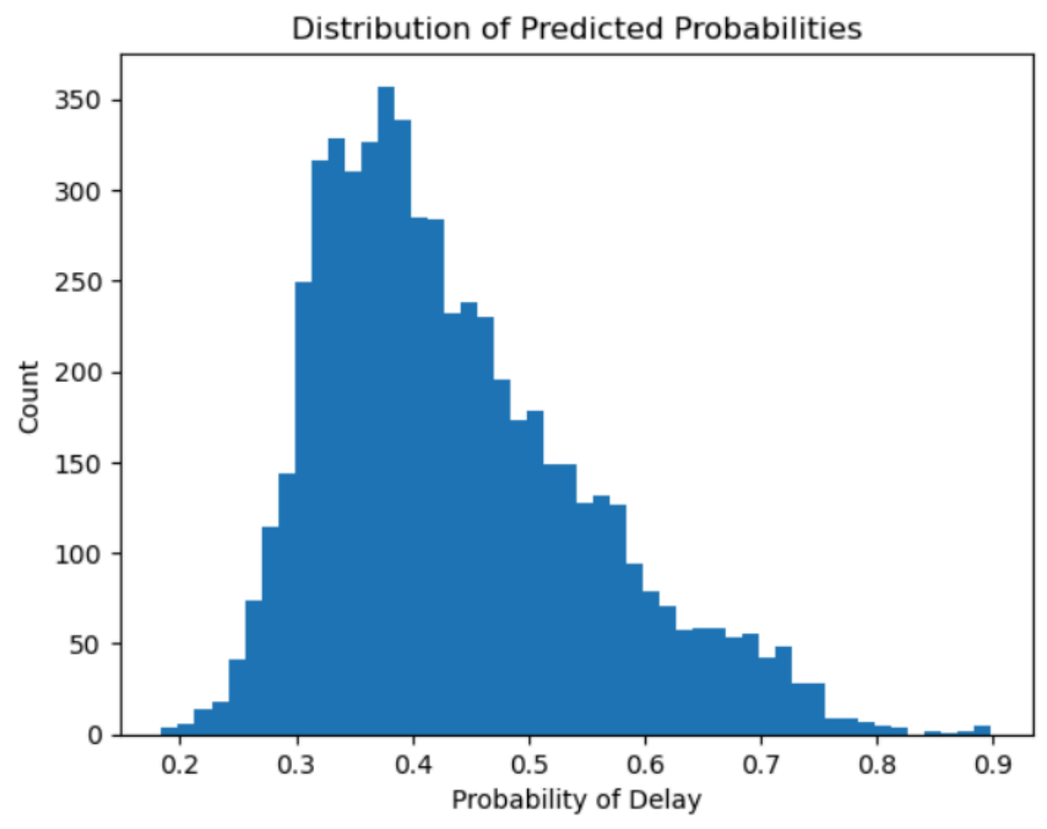
Delay flights: 17%



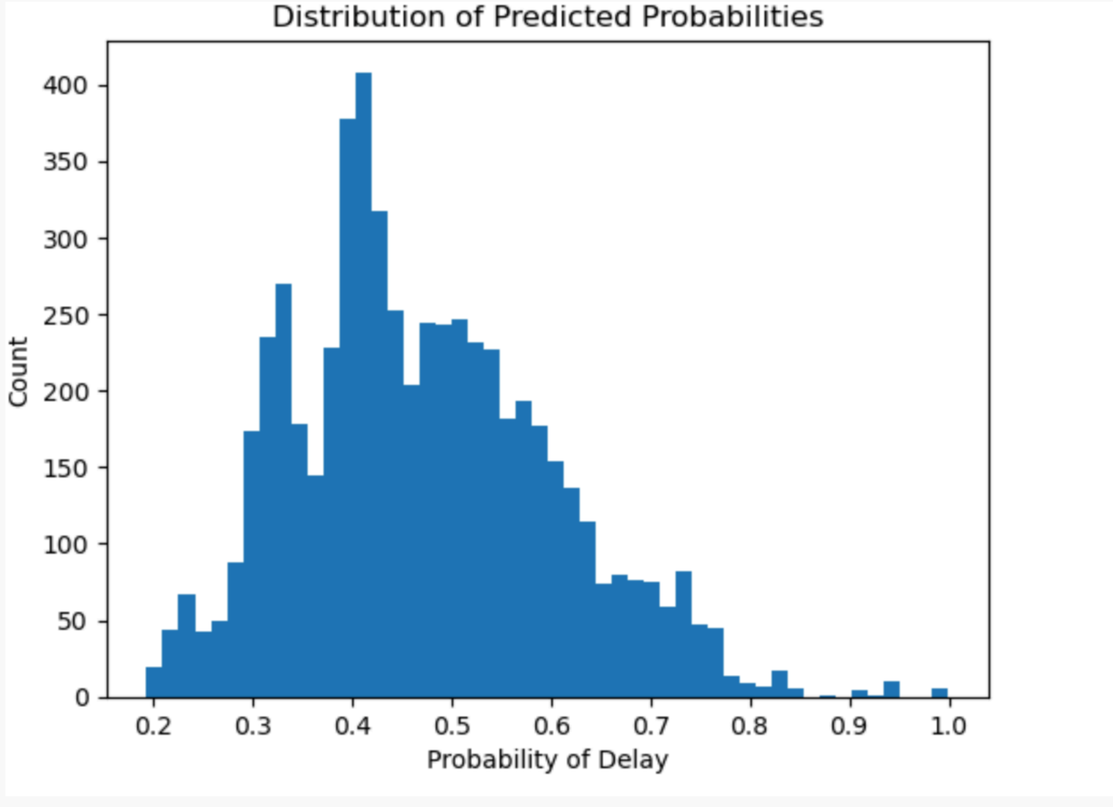


MODEL COMPARISON

Distribution of Predicted Probabilities



Random Forest



Logistic Regression

MODEL COMPARISON

Precision, Recall, F1 score

```
print(f"Best Threshold for F1-Score: {best_threshold:.4f}")
✓ 0.0s
Best Threshold for F1-Score: 0.4641
```

```
--- Performance at Threshold: 0.4641 ---
Precision: 0.2743
Recall:    0.5618
F1 Score:  0.3686
```

Random Forest

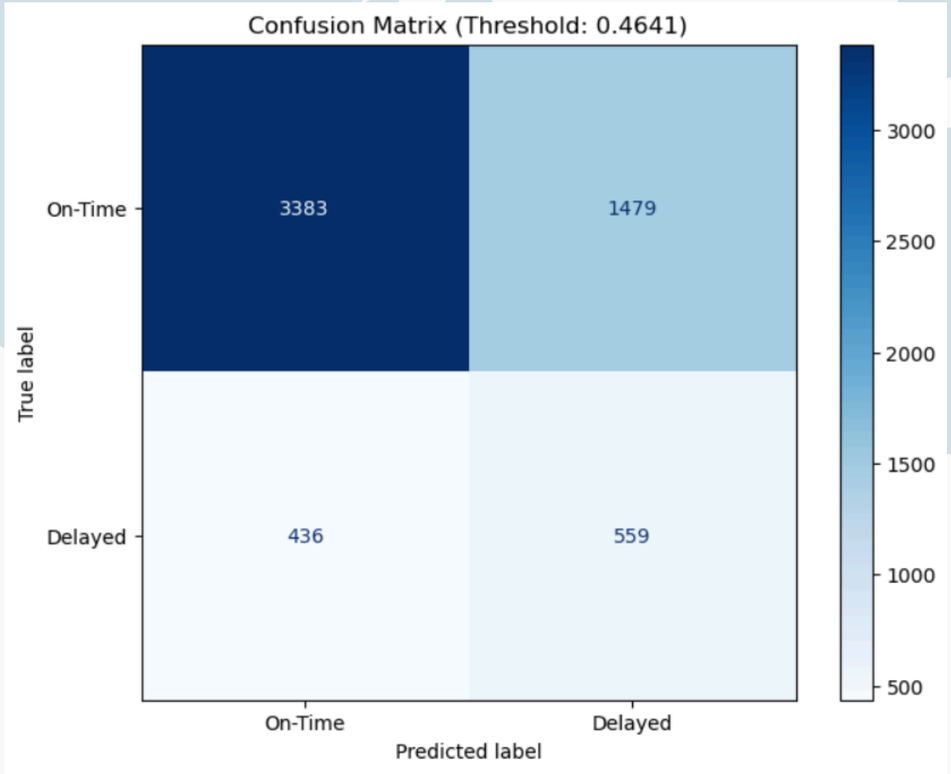
```
print(f"Best Threshold for F1-Score: {best_threshold:.4f}")
Best Threshold for F1-Score: 0.4836
```

```
--- Performance at Threshold: 0.4836 ---
Precision: 0.2417
Recall:    0.6111
F1 Score:  0.3463
```

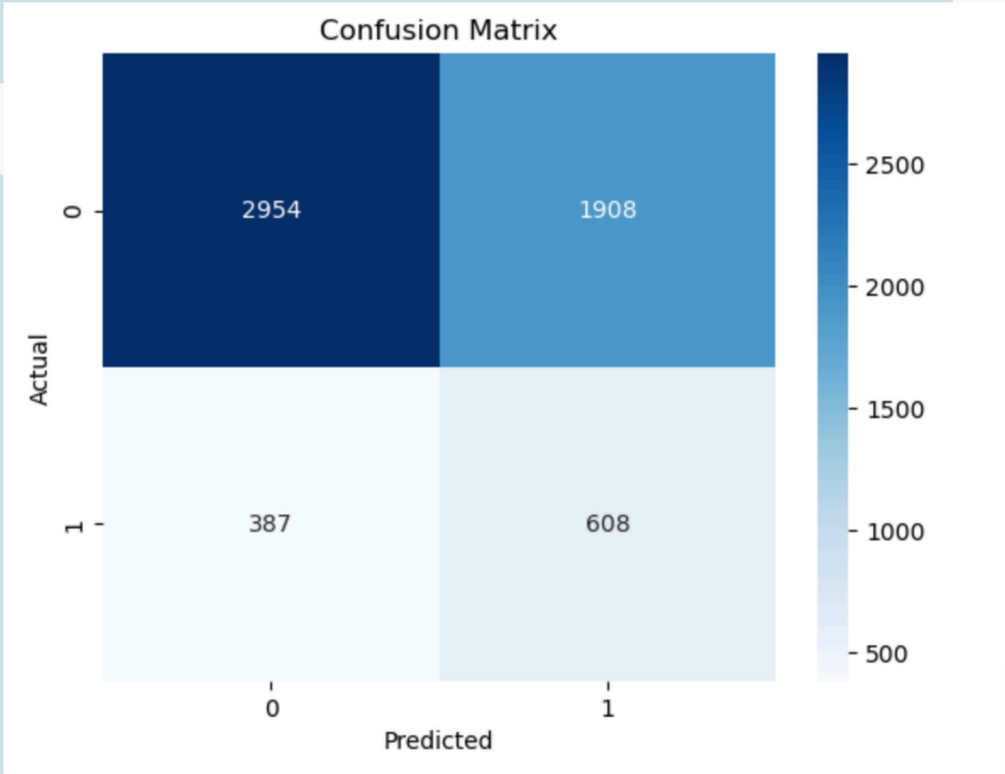
Logistic Regression

MODEL COMPARISON

Confusion Matricies



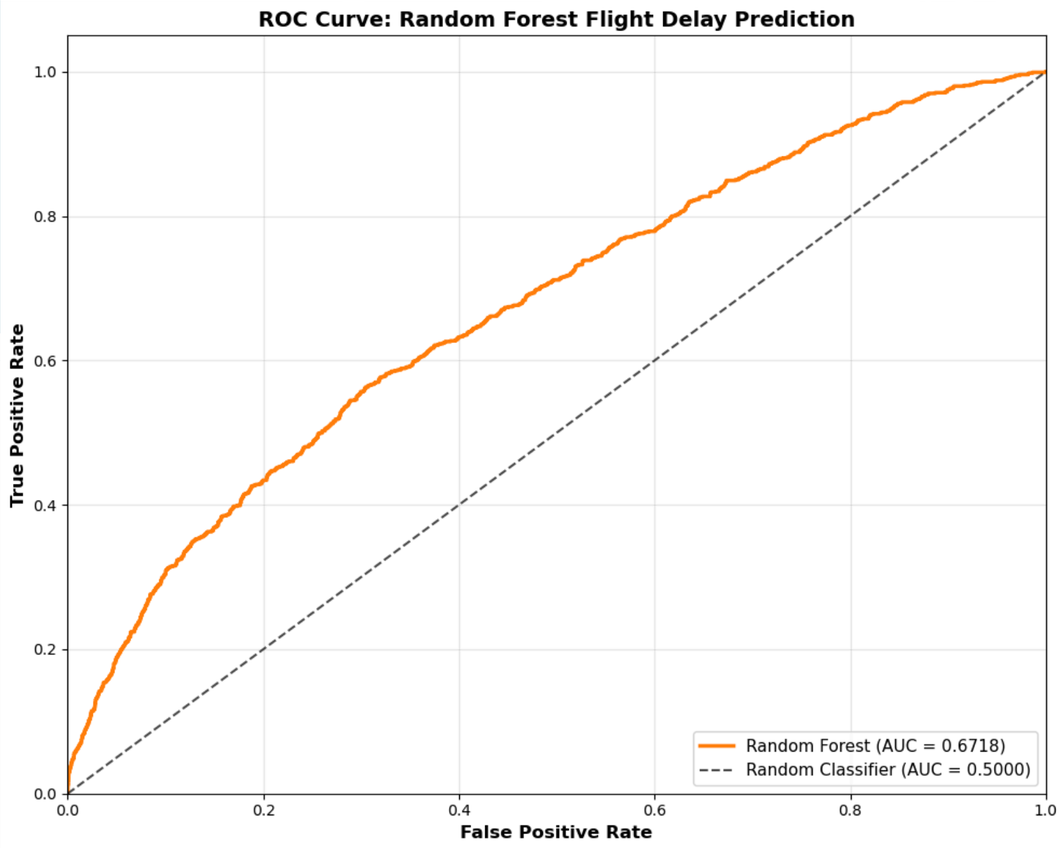
Random Forest



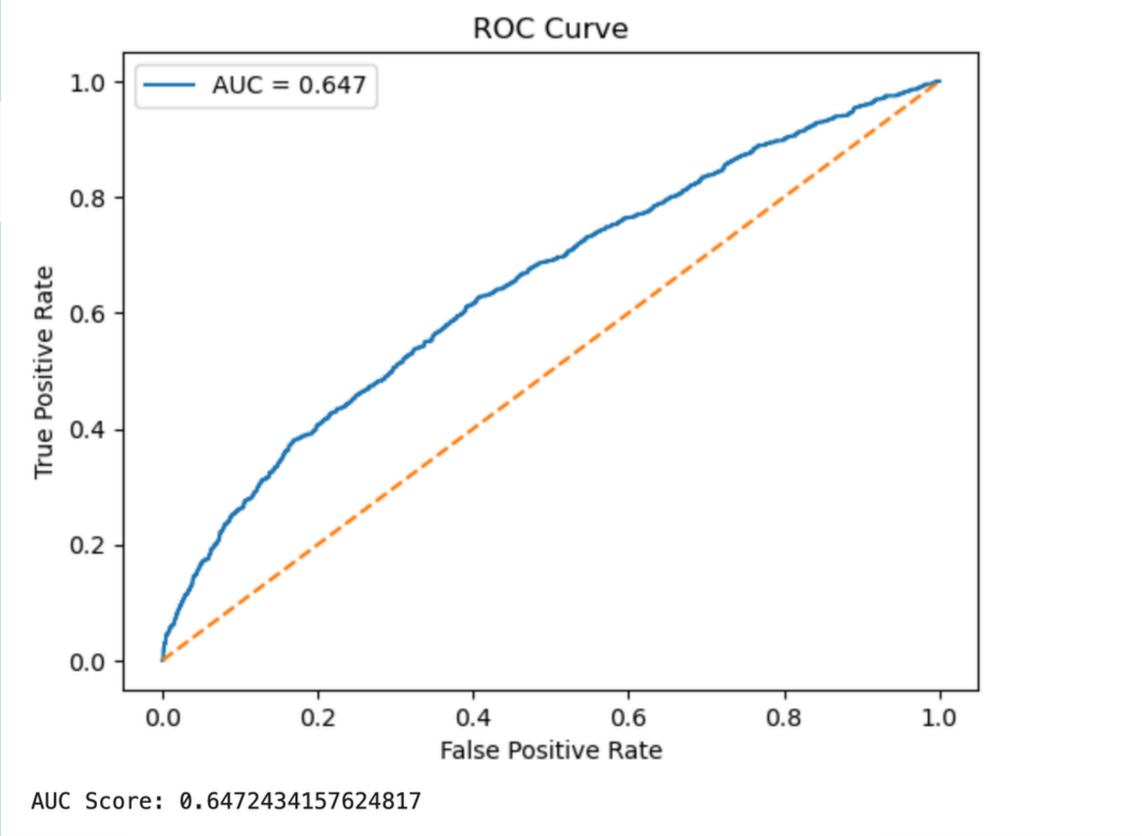
Logistic Regression

MODEL COMPARISON

ROC Curve



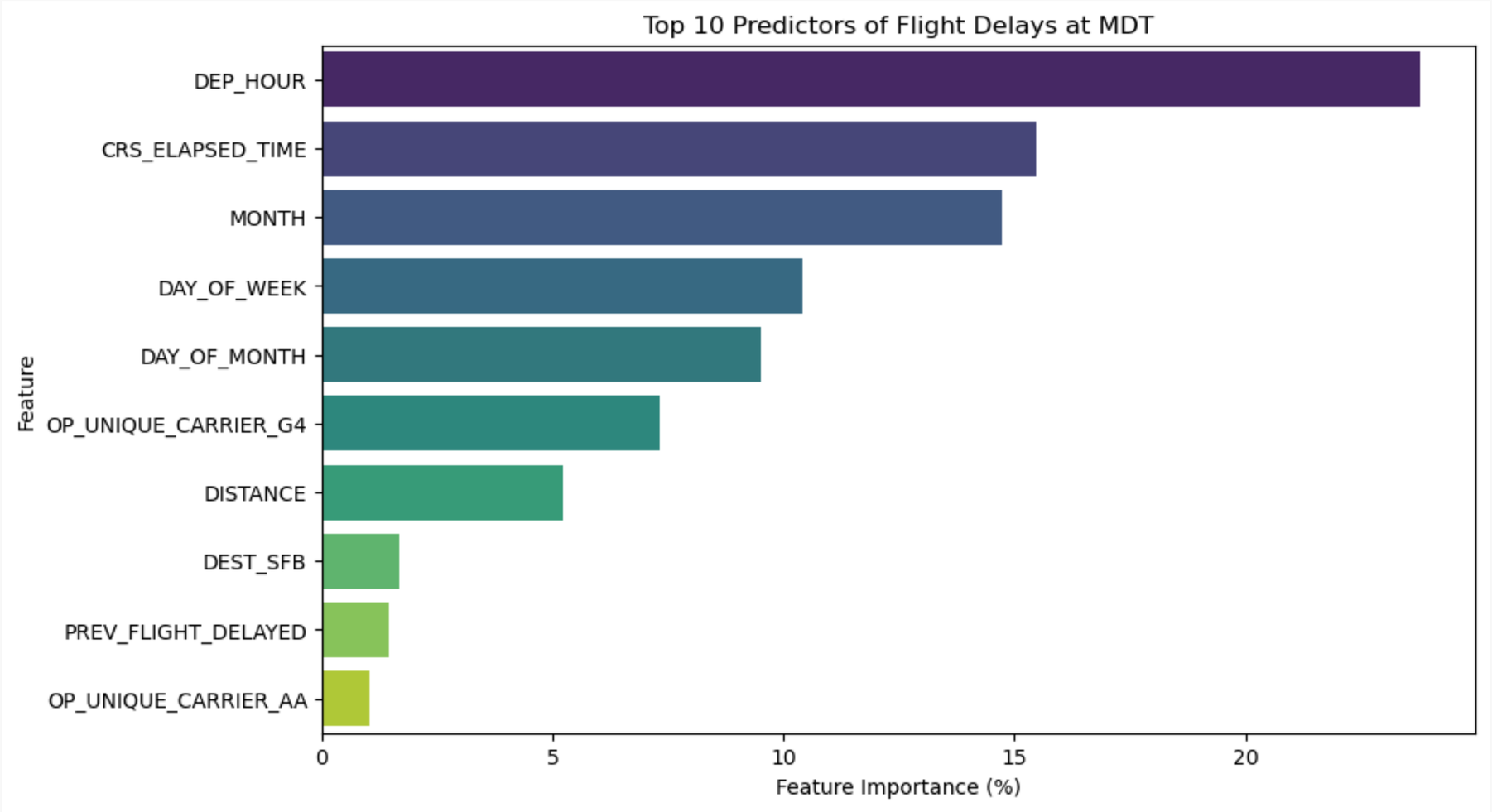
Random Forest



Logistic Regression

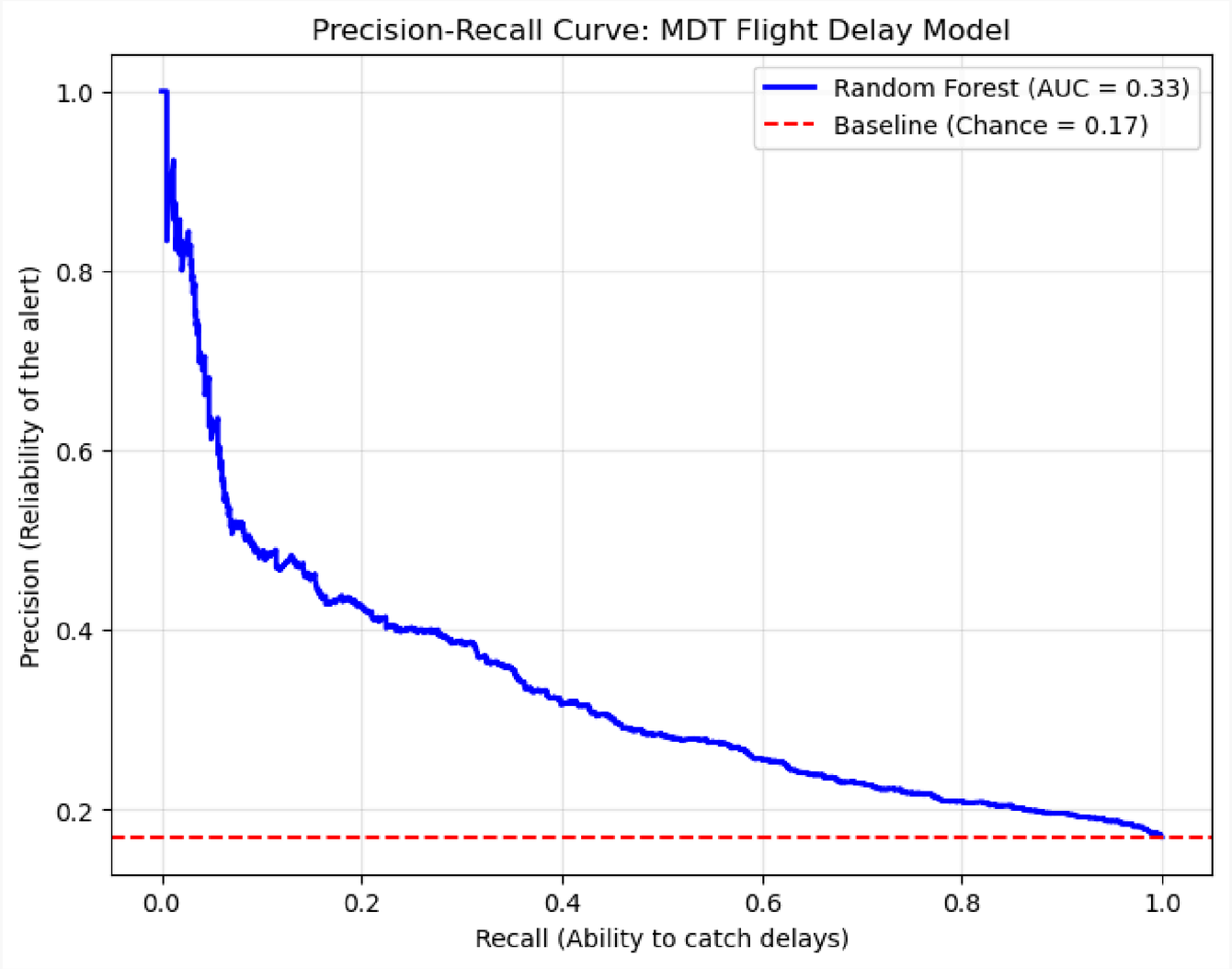
FEATURES IMPORTANCE

Random Forest



PRECISION VS. RECALL CURVE

Random Forest



LIMITATIONS



LIMITED FEATURE SET :	Does not include weather, airport congestion, or air traffic conditions
CLASS IMBALANCE :	Fewer delayed flights than on-time flights Model tends to over-predict delays when optimizing for recall
MDT-ONLY DATASET:	Results may not generalize to larger or busier airports
DATA GRANULARITY:	Lacks detailed operational data Limits ability to fully capture delay propagation mechanisms





FUTURE WORK

Incorporate additional data sources
Weather data (e.g., storms, visibility, precipitation)
Airport congestion and traffic volume

Expand dataset scope
Include multiple airports to improve generalizability
Capture network-wide delay propagation

Integrate advanced models
Gradient boosting (e.g., XGBoost, LightGBM)
Better handling of nonlinear relationships and imbalance

Enhance feature engineering
More detailed time features (peak hours, seasonal trends)
Improved operational features beyond previous delay (airline historical delay rate)



THANK YOU!

